

Quick Activity 2: Canteen Group Order

1. Problem Analysis

The program will calculate the total cost of a group order in the canteen by multiplying the quantity of each food item by its unit price, then adding all subtotals.

The program should determine:

- The cost of each ordered item
- The total bill for the group

2. Inputs – Process – Outputs

Input	Process	Output
Quantity and price of burgers	Multiply quantity x price	Burger subtotal
Quantity and price of drinks	Multiply quantity x price	Drinks subtotal
Quantity and price of fries	Multiply quantity x price	Fries subtotal
All Subtotals	Add all Subtotals	Total Cost

Formula:

- $\text{Subtotal} = \text{Quantity} \times \text{Unit Price}$
- $\text{Total Cost} = \text{Sum of all Subtotals}$

3. Decomposition

The problem can be divided into these steps:

1. Get the quantity and price of each food item
2. Compute subtotal for each item
3. Add all subtotals
4. Display the total bill

4. Pseudocode

START

INPUT burger_quantity

INPUT burger_price

INPUT drinks_quantity

INPUT drinks_price

INPUT fries_quantity

INPUT fries_price

burger_total = burger_quantity $\times$ burger_price

drinks_total = drinks_quantity $\times$ drinks_price

fries_total = fries_quantity $\times$ fries_price

total_cost = burger_total + drinks_total + fries_total

DISPLAY burger_total

DISPLAY drinks_total

DISPLAY fries_total

DISPLAY total_cost

END

5. Dry Run — Test Case 1

Given:

- 2 burgers $\times$ ₱50 = ₱100
- 3 drinks $\times$ ₱20 = ₱60
- 1 fries $\times$ ₱30 = ₱30

Calculation:

Item	Quantity	Unit Price	Subtotal
Burger	2	₱50	₱100
Drinks	3	₱20	₱60
Fries	1	₱30	₱30

Total Cost = ₱100 + ₱60 + ₱30 = ₱190

Output:

Total cost: ₱190

6. Dry Run — Test Case 2

Given:

- 4 burgers $\times$ ₱50 = ₱200
- 5 drinks $\times$ ₱20 = ₱100
- 2 fries $\times$ ₱30 = ₱60

Calculation:

Item	Quantity	Unit Price	Subtotal
Burger	4	₱50	₱200
Drinks	5	₱20	₱100
Fries	2	₱30	₱60

Total Cost = ₱200 + ₱100 + ₱60 = ₱360

Output:

Total cost: ₱360

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